PHAETON CAPITAL

PRODUCT ROADMAP — V2

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Scope: P0–P3 Feature Roadmap incl. Cultural Arbitrage Assessment

CLASSIFICATION: INTERNAL

1. EXECUTIVE SUMMARY

This document presents the second-generation product roadmap for Phaeton Capital following the completion of all P0–P3 features from the initial assessment. Four new ideas were evaluated through rigorous research across academic finance literature, API availability, engineering complexity, and expected alpha generation. The findings are direct: two ideas are high-value quick wins, one requires architectural changes but is genuinely differentiated, one is intellectually seductive but empirically unsupported in its original form — and has been redirected into a stronger variant.

Ideas evaluated:

- Cultural dimension / cultural arbitrage trading signals
- Smart search bar: company name → ticker autocomplete
- Multi-timeframe recommendations (15d / 30d / 90d)
- Google Trends integration + Polymarket prediction markets

2. HONEST IDEA ASSESSMENTS

2.1 Cultural Arbitrage — Redirected

Original idea: Use Hofstede cultural dimensions to detect tradeable signals across markets based on how different cultures process the same financial information.

VERDICT: THEORETICALLY APPEALING — EMPIRICALLY WEAK

Academic research on Hofstede dimensions (Uncertainty Avoidance, Individualism/Collectivism, Long-Term Orientation, Power Distance) and equity market behaviour exists but is severely limited. The seminal study most cited — Chui, Titman & Wei (2010) — shows individualism scores predict momentum profitability cross-nationally. However, every apparent cultural effect in the literature dissolves when controlling for regulatory regime, market maturity, institutional ownership concentration, and economic development level. This is not a minor caveat — it means the cultural signal is likely a **proxy variable** for institutional quality, not culture itself.

Why it fails as a trading signal:

- Hofstede scores are static or change over decades — markets change daily. Temporal mismatch makes it untradeable.
- No consistent out-of-sample alpha has been documented in 20+ years of academic attempts.
- The A/H share premium (Shanghai vs Hong Kong) — the most cited real-world example — is fully explained by capital controls, liquidity, and index inclusion. Not culture.
- If real and stable, it would have been systematized and arbitraged away long ago.
- Academic publication bias: studies finding 'no effect' go unpublished.

What to build instead — the real version of this idea:

The spirit of cultural arbitrage is correct: different investor communities price the same stock differently. The mistake is using static Hofstede theory instead of live behavioural data. Two concrete, measurable alternatives exist:

- **Regional Sentiment Divergence (P3):** Scrape the same ticker across regional Reddit communities (r/AusFinance, r/UKInvesting, r/EUstocks). When US sentiment is +0.4 and EU is –0.2 for the same stock, that divergence is real, observable, and potentially tradeable. Your app is already positioned to do this.
- **ADR / Cross-Listing Gap Monitor (P2):** ASML, Toyota, Sony, SAP, HSBC all trade on both their home exchange and US markets. Real price gaps exist due to timezone asymmetry and liquidity imbalances — this is mechanical arbitrage, directly observable via yfinance.
- **Cultural Context Card (P2, static):** Non-trading enrichment. Map company's home country to Hofstede scores, render a plain-English implication card. Zero API calls, 4h build, genuine context for international stocks.

2.2 Smart Search Bar — P0 Quick Win

VERDICT: BUILD THIS FIRST. HIGHEST IMPACT / LOWEST EFFORT CHANGE IN THE BACKLOG.

The current UX requires users to know the exact ticker symbol. This is fatal to adoption. A normal person types 'Apple' or 'Volkswagen' and gets nothing. The fix is trivial and the infrastructure already exists.

Technical approach:

- Yahoo Finance undocumented search: query2.finance.yahoo.com/v1/finance/search?q={query} — free, stable, used by every major finance tool. Returns ticker, name, exchange in ~50ms.
- Financial Modeling Prep official fallback: /v3/search?query={q}&apikey;=... — 250 req/day free tier, enough for personal use.
- New Next.js route: GET /api/search?q={query} — proxies to Yahoo Finance, filters to EQUITY type, returns top 6 results.
- Command palette update: debounced keystroke (300ms) → fetch → dropdown replaces static popular tickers list.
- On select: triggerScrape(symbol) directly, close palette.

2.3 Multi-Timeframe Recommendations — P1 High Value

VERDICT: NECESSARY. THE CURRENT 15-DAY-ONLY HORIZON IS SILENTLY WRONG FOR MOST USERS.

Every signal, score, and prediction the app generates is calibrated to a 15-day holding period. This is never communicated to the user. A day trader and a pension fund manager both see 'BUY' and interpret it through completely different lenses. The system needs to be explicit about its horizon and ideally offer multiple.

Implementation strategy:

- Schema: Add horizon Int @default(15) to RecommendationScore, making the compound key @@unique([ticker, horizon]).
- Logic: computeRecommendation(ticker, horizon) accepts 15 | 30 | 90. Horizon affects Kelly lookback window, prediction resolution threshold, and backtest holding period.
- Worker: Run all three horizons per cycle. Adds ~3s per ticker — acceptable.
- UI: Three tab pills in hero section (15D / 30D / 90D). State-controlled toggle. Scores will diverge meaningfully — a stock oversold short-term may be weak at 90d.
- Immediate fix (1h): Add explicit horizon labels to every signal, backtest result, and prediction table. No logic change needed.

2.4 Google Trends — P1. Polymarket — Skip.

VERDICT: GOOGLE TRENDS = REAL ALPHA. POLYMARKET = WRONG TOOL FOR THIS JOB.

Google Trends:

Preis, Moat & Stanley (2013, Nature Scientific Reports) demonstrated that Google search volume for financial keywords preceded market movements. Subsequent research at UCL confirmed search interest for '{ticker} stock' predicts 2-week forward returns with statistical significance. This is not hype — it's peer-reviewed with replication.

- pytrends library (unofficial wrapper): moderate reliability, rate-limited to ~1 req/5s before IP block.
- Data resolution: weekly (daily available for recent 3 months only). 1-2 day data lag.
- Best keywords: '{ticker} stock', 'buy {ticker}', '{company} earnings' — these outperform generic company name searches.
- Engineering risk: unofficial API, no SLA. Mitigate with aggressive caching (6h TTL) and graceful fallback.
- Signal: z-score of last week vs 12-week average. trends_score > 70 → +5 to sentimentScore.

Polymarket:

Polymarket has a REST API but equity-specific prediction markets are nearly nonexistent. The platform focuses on political events, sports, and macro predictions. Volume on individual stock markets is too thin to produce reliable probability signals. **Skip entirely** unless the app pivots toward event-driven macro trading.

3. MASTER ROADMAP TABLE

Priority	Feature	Effort	Impact	Status
P0 NOW	Smart search: company name → ticker autocomplete	3h	★★★★★	Not started
P0 NOW	Explicit timeframe labels on all signals (15d clarity)	1h	★★★★■	Not started
P1 NEXT	Multi-horizon recommendations (15d / 30d / 90d tabs)	1 day	★★★★★	Not started
P1 NEXT	Google Trends signal + sparkline chart	1 day	★★★★■	Not started
P2	ADR / Cross-listing price gap monitor	2 days	★★★■■	Not started
P2	Cultural context card (static Hofstede data)	4h	★★★■■	Not started
P3	Regional sentiment divergence (multi-community scraping)	3 days	★★★★■	Not started
SKIP	Polymarket integration	—	★■■■■	Rejected
SKIP	Hofstede arbitrage trading signals	—	★■■■■	Rejected

★★★★★ = transformative ★★★★■ = high ★★★■■ = medium ★★■■■ = low ★■■■■ = negligible

4. DETAILED IMPLEMENTATION SPECS

4.1 P0-A: Smart Search Bar

Files to create/edit:

- NEW: apps/web/app/api/search/route.ts
- EDIT: apps/web/app/DashboardClient.tsx (command palette)

apps/web/app/api/search/route.ts:

GET /api/search?q={query} handler. Fetch from Yahoo Finance search endpoint with 5s timeout. Filter results to quoteType === 'EQUITY'. Return array of { symbol, shortname, exchange }. Add Financial Modeling Prep as fallback if Yahoo fails. Cache results for 60s with simple Map.

DashboardClient.tsx — command palette changes:

- Add searchResults state: useState<{symbol,shortname,exchange}[]>([])
- Add useEffect debouncing searchQuery → fetch /api/search?q={query} after 300ms
- Replace static Popular Tickers buttons with dynamic dropdown list
- On item click: triggerScrape(item.symbol), setIsCommandOpen(false)
- Show exchange badge next to ticker (NYSE, NASDAQ, LSE etc.)

4.2 P0-B: Timeframe Clarity Labels

Copy-only changes. No logic modifications required. Every place the system shows a signal, score, prediction, or backtest result must explicitly state its time horizon.

- Hero section signal word: add '15-Day Outlook' sub-label below STRONG BUY / BUY / etc.
- Prediction audit trail table: rename column 'Outcome' → 'Outcome (15d)'
- Backtest stats card: add sub-label '2-Year Walk-Forward · 15-Day Holds'
- Score badge in hero: 'Score: 74.2/100' → 'Score: 74.2/100 · 15d'

4.3 P1-A: Multi-Horizon Recommendations

Schema changes (packages/db/prisma/schema.prisma):

- Add horizon Int @default(15) field to RecommendationScore model
- Change unique constraint from @@unique([ticker]) to @@unique([ticker, horizon])
- Add horizon Int @default(15) field to PredictionRecord model

Worker changes (apps/worker/src/recommendation.ts):

- computeRecommendation(ticker, horizon: 15 | 30 | 90 = 15)
- Pass horizon to prisma upsert/create calls
- Adjust Kelly lookback: horizon * 2 trading days
- Adjust prediction resolution window to match horizon

Worker orchestration (apps/worker/src/index.ts):

- After existing computeRecommendation(ticker), add calls for 30 and 90 day horizons
- Wrap in Promise.allSettled to avoid one horizon failure blocking others

Frontend changes:

- page.tsx: fetch all three horizons from DB, pass as props
- DashboardClient.tsx: add selectedHorizon state (15 | 30 | 90), tab pill UI in hero, derive displayed rec from props based on selected horizon

4.4 P1-B: Google Trends Signal

Python worker (apps/python_worker/main.py):

- pip install pytrends — add to requirements.txt
- New GET /trends/{ticker} endpoint
- build_payload([f'{ticker} stock'], timeframe='today 3-m', geo='US')
- Compute z-score: (last_week_interest - 12w_mean) / 12w_std
- trends_score = clamp(50 + z_score * 10, 0, 100)
- time.sleep(10) between calls — mandatory to avoid IP block
- Return: { trends_score, weekly_data: [{week, interest}], direction: 'up'|'down'|'flat' }

Schema changes:

- Add trends_score Float? to QuantMetrics model
- New TrendsHistory model: id, ticker, week_start Date, interest Int, createdAt DateTime

Worker (apps/worker/src/trends.ts — new file):

- fetchGoogleTrends(ticker): calls python_worker /trends/{ticker} with 30s timeout
- Upsert trends_score into QuantMetrics
- Batch insert TrendsHistory rows (skip if already exists for that week)

Recommendation engine nudge:

- If trends_score > 70: sentimentScore = clamp(sentimentScore + 5)
- If trends_score < 30: sentimentScore = clamp(sentimentScore - 5)

UI (DashboardClient.tsx):

- Add Google Trends mini-section inside Quant Models card

- 12-week sparkline using Recharts AreaChart, color-coded by direction
- Single stat tile: 'Search Momentum' with trends_score value

4.5 P2-A: ADR / Cross-Listing Gap Monitor

Tracks real-time price discrepancy between a stock's home-market listing and its US ADR. Examples: ASML (AEX + NASDAQ), Toyota (TSE + NYSE), SAP (XETRA + NYSE).

- Python endpoint GET /crosslist/{ticker}: fetch both prices via yfinance, apply FX conversion, compute gap_pct
- New CrossListingData Prisma model: ticker, home_ticker, home_exchange, home_price, adr_price, fx_rate, gap_pct, updatedAt
- Worker: scrape cross-listing data after fundamentals sync
- UI: Conditional card in Fundamentals section. If |gap_pct| > 0.5%: show amber alert 'Cross-listing discrepancy: +1.2% (NASDAQ premium vs AEX)'

4.6 P2-B: Cultural Context Card

Static enrichment feature. No trading signal — pure editorial context. Maps company's listed exchange country to Hofstede dimension scores and renders plain-English market behaviour implications.

- Static JSON bundled in worker: Map for 50 major markets
- Worker resolves exchange country from fundamentalData.exchange field
- Stores country_code and cultural_profile JSON string in MacroIndicators or new field
- UI: Single collapsible card in Macro section. Shows 4 dimension bars (UAI, IDV, LTO, PDI) with 1-sentence implication each
- Example: 'Netherlands — Uncertainty Avoidance: 53/100. Implication: moderate sell-off response to ambiguous data vs. high-UAI markets like Japan (92/100).'

4.7 P3: Regional Sentiment Divergence

The real version of cultural arbitrage. Expands the Reddit scraper to pull from regional finance communities for the same ticker and detects when communities price it differently.

- Schema: Add region String @default('US') field to Sentiment model
- Scraper: For each ticker, scrape r/investing (US), r/AusFinance (AU), r/UKInvesting (UK), r/EUstocks (EU)
- New divergence_score computed in recommendation.ts: max regional sentiment gap
- Worker: After sentiment batch, compute regional means and store in new RegionalSentiment table
- UI: Radar/spider chart showing sentiment by region. Alert if divergence > 0.3: 'Regional Divergence Detected: US +0.4 vs EU -0.2'

5. RECOMMENDED IMPLEMENTATION ORDER

The following order maximises user-visible impact at each step while keeping architectural risk low. P0 items can be done in a single session. P1 items require schema migrations.

Step	Feature	Rationale	Est.
1	Smart search autocomplete	Makes app usable for any normal person — immediate adoption impact	3h
2	Timeframe labels	Zero logic risk, removes a silent UX lie that misleads every user	1h
3	Google Trends signal	Genuine data edge, differentiates from every other retail tool	1d
4	Multi-horizon recs	Requires schema migration — do after Trends to batch the DB push	1d
5	Cultural context card	Static data, zero risk, adds polish for international stocks	4h
6	ADR gap monitor	Real mechanical alpha signal — needs yfinance dual-ticker logic	2d
7	Regional sentiment divergence	Biggest architectural change — expand scraper to 4 communities	3d

6. WHAT NOT TO BUILD — AND WHY

Product discipline requires saying no. The following were evaluated and rejected.

Feature	Why Rejected
Hofstede arbitrage trading signals	No consistent out-of-sample alpha in 20+ years of academic research. Every apparent cultural effect dissolves when controlling for institutional quality and regulation. Building this would create false credibility for a signal that doesn't work.
Polymarket integration	Prediction market data for individual equities is nearly nonexistent on Polymarket. The platform is political/sports/macro focused. Equity volume is too thin for reliable probability signals. Would add complexity with no measurable alpha.

7. CURRENT STACK STATUS (POST P0–P3)

All original P0–P3 items from the first assessment have been completed. The following is the current feature state entering this roadmap.

Component	Feature	Status
Python Worker	Options flow (put/call ratio, max pain, IV percentile)	DONE ✓
Python Worker	Walk-forward backtest (RSI+MACD+SMA, 15d holds)	DONE ✓
Python Worker	GARCH, HMM, Monte Carlo, Hurst, Kelly, Bayesian	DONE ✓
Worker (TS)	Regime-aware weights (CRISIS/BULL/BEAR/NEUTRAL)	DONE ✓
Worker (TS)	Ridge regression weight optimisation from prediction history	DONE ✓
Worker (TS)	Options flow scraping + DB persistence	DONE ✓
Worker (TS)	SSE broadcast on sentiment/quant/recommendation events	DONE ✓
Worker (TS)	LLM narrative generation via Gemini (async, non-blocking)	DONE ✓
Web (Next.js)	SSE client replacing 30s polling	DONE ✓
Web (Next.js)	Prediction audit trail UI (stats bar + table)	DONE ✓
Web (Next.js)	LLM narrative display in hero section	DONE ✓
Web (Next.js)	Backtest API route (/api/backtest)	DONE ✓
Schema	OptionsFlow model, narrative field on RecommendationScore	DONE ✓

End of Document — Phaeton Capital Roadmap V2 — 29 March 2026